

20260910_p1

September 12, 2026

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[1]: import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

[2]: df_cdmx1 = pd.read_csv('./data/CDMX_2015_2020.csv')
df_cdmx2 = pd.read_csv('./data/CDMX_2020_2025.csv')

df_cuerna1 = pd.read_csv('./data/CUERNA_2015_2020.csv')
df_cuerna2 = pd.read_csv('./data/CUERNA_2020_2025.csv')

df_puebla1 = pd.read_csv('./data/PUEBLA_2015_2020.csv')
df_puebla2 = pd.read_csv('./data/PUEBLA_2020_2025.csv')

[3]: df_cdmx1 = df_cdmx1[df_cdmx1['DATE'] < '2020-01-01']
df_cuerna1 = df_cuerna1[df_cuerna1['DATE'] < '2020-01-01']
df_puebla1 = df_puebla1[df_puebla1['DATE'] < '2020-01-01']

[4]: df_general = pd.concat([df_cdmx1, df_cdmx2,
                             df_cuerna1, df_cuerna2,
                             df_puebla1, df_puebla2]).sort_values(by = 'DATE')

[5]: df_general = df_general.reset_index(drop = True)

[6]: df_general.drop(['TMAX_ATTRIBUTES', 'TMIN_ATTRIBUTES'], axis = 1, inplace =
↳ True)

[7]: df_general['TMAX'] = (df_general['TMAX'] - 32)/1.8
df_general['TMIN'] = (df_general['TMIN'] - 32)/1.8

[8]: df_general['YEAR'] = df_general['DATE'].apply(lambda date: date.split(sep =
↳ '-') [0])
df_general['MONTH'] = df_general['DATE'].apply(lambda date: date.split(sep =
↳ '-') [1])
df_general['DAY'] = df_general['DATE'].apply(lambda date: date.split(sep =
↳ '-') [2])

[9]: df_general.dropna(axis = 0, inplace = True)
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[10]: df_general
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[10]:
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	STATION	NAME	DATE	TMAX \
1	MXM00076683	TLAXCALA DE XICONTECATL DGE, MX	2015-01-01	23.888889
3	MXM00076683	TLAXCALA DE XICONTECATL DGE, MX	2015-01-02	22.222222
4	MXM00076680	MEXICO CITY, MX	2015-01-02	22.222222
5	MXM00076726	CUERNAVACA, MX	2015-01-02	27.222222
8	MXM00076683	TLAXCALA DE XICONTECATL DGE, MX	2015-01-03	22.222222
...	...	...	...	...
14855	MXM00076683	TLAXCALA DE XICONTECATL DGE, MX	2025-11-27	22.222222
14856	MXM00076680	MEXICO CITY, MX	2025-11-28	17.777778
14857	MXM00076683	TLAXCALA DE XICONTECATL DGE, MX	2025-11-28	21.111111
14858	MXM00076683	TLAXCALA DE XICONTECATL DGE, MX	2025-11-29	21.111111
14859	MXM00076680	MEXICO CITY, MX	2025-11-30	22.222222

	TMIN	YEAR	MONTH	DAY
1	4.444444	2015	01	01
3	4.444444	2015	01	02
4	8.333333	2015	01	02
5	12.777778	2015	01	02
8	4.444444	2015	01	03
...	...	...	...	...
14855	10.555556	2025	11	27
14856	11.111111	2025	11	28
14857	8.888889	2025	11	28
14858	6.666667	2025	11	29
14859	11.666667	2025	11	30

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[7402 rows x 8 columns]
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[11]: df_general_tmin = df_general.drop(['TMAX'], axis = 1)
df_general_tmax = df_general.drop(['TMIN'], axis = 1)
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[12]: df_tmax = df_general_tmax.groupby(['YEAR', 'MONTH', 'DAY'])
df_tmin = df_general_tmin.groupby(['YEAR', 'MONTH', 'DAY'])
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[13]: def per_year():
    years = [str(y) for y in range(2015, 2026, 1)]
    year = input("Enter a year from 2015 to 2025 (leave blank for all years): ")
    year = year.strip()

    if year and year not in years:
        print("Invalid year")
        return

    if year:
        fig, ax = plt.subplots(figsize = (16, 6))
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axes = [ax]
years_to_plot = [year]
filename = 'temperaturas_' + year + '.png'
else:
    fig, ax = plt.subplots(len(years), 1, figsize = (16, 40))
    axes = ax
    years_to_plot = years
    filename = 'temperaturas_2015_a_2025.png'

for i in range(len(years_to_plot)):
    current_year = years_to_plot[i]
    tmax = np.array(df_tmax.max().loc[current_year]['TMAX'])
    tmin = np.array(df_tmin.min().loc[current_year]['TMIN'])
    locs = np.linspace(0, len(tmax) - 1, 12)

    axes[i].plot(tmin, label = '$T_{min}$')
    axes[i].plot(tmax, label = '$T_{max}$')
    axes[i].grid(alpha = 0.3)
    axes[i].legend()
    axes[i].set_title(current_year)
    axes[i].set_xticks(locs)
    axes[i].set_xticklabels(['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec'])
    axes[i].set_ylabel('T [°C]')

plt.tight_layout()
plt.savefig(filename, dpi = 300, bbox_inches = 'tight')
plt.show()

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[14]: def yearly_avg():
    years = [str(y) for y in range(2015, 2026, 1)]
    tmax_avg = []
    tmin_avg = []

    for i in range(len(years)):
        year = years[i]
        tmax = np.array(df_tmax.max().loc[year]['TMAX'])
        tmin = np.array(df_tmin.min().loc[year]['TMIN'])
        tmax_avg.append(tmax.mean())
        tmin_avg.append(tmin.mean())

    flag = input("Choose: \n"
                 "[1] for bar graph \n"
                 "[2] for line graph \n")

    plt.figure(figsize = (12, 6))
    if flag == '1':

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x = np.arange(len(years))
width = 0.4
plt.bar(x - width/2, tmax_avg, width, label = '$T_{max}$')
plt.bar(x + width/2, tmin_avg, width, label = '$T_{min}$')
plt.xticks(x, years)
elif flag == '2':
    plt.plot(years, tmax_avg, marker = 'o', label = '$T_{max}$')
    plt.plot(years, tmin_avg, marker = 'o', label = '$T_{min}$')
    plt.xticks(years)
else:
    plt.close()
    print("Invalid choice")
    return

plt.ylabel('T [°C]')
plt.grid(alpha = 0.3)
plt.legend()
plt.tight_layout()
plt.savefig('temperaturas_promedio_2015_a_2025.png', dpi = 300, bbox_inches_
↳= 'tight')
plt.show()

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[15]: def records():
    dates = []
    maximum = []
    minimum = []
    max_breaks = []
    min_breaks = []

    for month in range(1, 13):
        for day in range(1, 32):
            data = df_general[
                (df_general['MONTH'] == str(month).zfill(2))
                & (df_general['DAY'] == str(day).zfill(2))
            ]
            old = data[
                (data['YEAR'] >= '2015')
                & (data['YEAR'] <= '2024')
            ]
            new = data[data['YEAR'] == '2025']

            if len(old):
                dates.append(str(month) + '-' + str(day))
                maximum.append(old['TMAX'].max())
                minimum.append(old['TMIN'].min())

                if len(new) and new['TMAX'].max() > maximum[-1]:

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        max_breaks.append(new['TMAX'].max())
    else:
        max_breaks.append(np.nan)

    if len(new) and new['TMIN'].min() < minimum[-1]:
        min_breaks.append(new['TMIN'].min())
    else:
        min_breaks.append(np.nan)

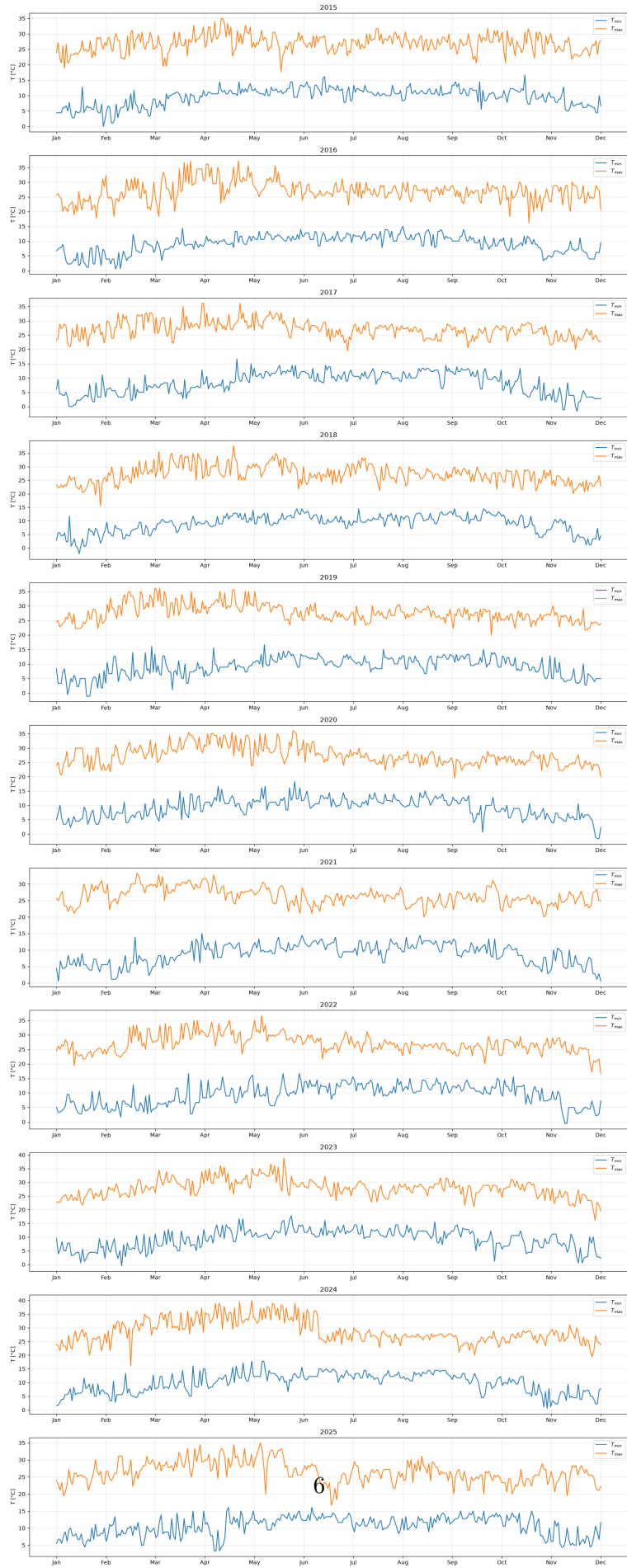
x = range(len(dates))
labels = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec']
month_positions = []
month_labels = []

for month in range(1, 13):
    for i in range(len(dates)):
        if dates[i].split('-')[0] == str(month):
            month_positions.append(i)
            month_labels.append(labels[month - 1])
            break

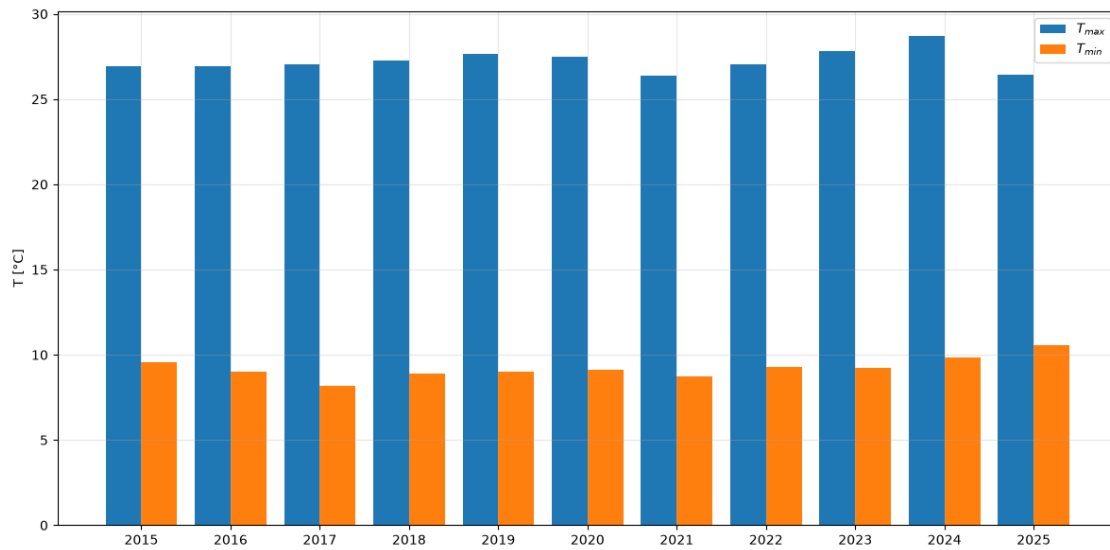
plt.figure(figsize = (18, 6))
plt.plot(x, minimum, alpha = 0.5, label = '$T_{min}$ record 2015-2024')
plt.plot(x, maximum, alpha = 0.5, label = '$T_{max}$ record 2015-2024')
plt.scatter(x, max_breaks, color = 'black', marker = 'o', label = '2025 max_
record')
plt.scatter(x, min_breaks, color = 'black', marker = 'v', label = '2025 min_
record')
plt.xticks(month_positions, month_labels)
plt.ylabel('T [°C]')
plt.grid(alpha = 0.3)
plt.legend()
plt.tight_layout()
plt.savefig('climatologia_record_2015_a_2025.png', dpi = 300, bbox_inches =
tight')
plt.show()

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[16]: per_year()



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[17]: yearly_avg()
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[18]: records()
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